[536709] Stochastic Processes

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3 Markov Process

We consider a stochastic process $\mathbf{X} = \{X_t\}_{t\in I}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ with state space $(S, \mathcal{B})$. In this section we consider *Markov process* and *Markov chain*, which have broadly and deeply studied. Most of stochastic processes we imagine would be the one whose present situation depends only on the previous status. For instance, the position of simple random walk with jump width 1 at time n denoted by S_n is $S_{n-1}+1$ or $S_{n-1}-1$. Of course, we can also say S_n should be $S_{n-2}+2$, S_{n-2} or $S_{n-2}-2$ but if the steps become larger, it is quite tough to express S_n. The simplest expression is by the previous step. In statistical mechanics, there are several polymer models. One of them, self-avoiding walk model is no longer a Markov process because the path has to memorize the whole steps due to self-avoidance. Form this point of view, the Markov process is the simplest case to learn first.

We recall that $I = \mathbb{Z}_+$ for discrete setting or $I = \mathbb{R}_+$ for continuous setting. We will treat the continuous setting in the next section but we give the definition of Markov process under a general setting.

Definition 3.1 (Markov process). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If a stochastic process $\mathbf{X} = \{X_t\}_{t\in I}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ with state space $(S, \mathcal{B})$ satisfies is called **Markov process** if and only if $\mathbf{X}$ is $\mathcal{F}_t$-adapted and for any $B \in \mathcal{B}$ and $s \le t$,

$$\mathbb{P}(X_t \in B \mid \mathcal{F}_s) = \mathbb{P}(X_t \in B \mid X_s) \qquad \mathbb{P}\text{-a.s.} \tag{3.1}$$

We remark here again that we treat $I = \mathbb{Z}_+$ for discrete models. For random walk $S = \{S_n\}_{n\in\mathbb{Z}_+}$, it satisfies

$$\mathbb{P}(S_{n+1} = y \mid S_0 = 0, S_1 = x_1, \cdots, S_n = x_n) = \mathbb{P}(S_{n+1} = y \mid S_n = x_n)$$

which means that if we know the information of the position at n-step: $S_n = x_n$, then the probability law of the position at $(n+1)$-step is not related to positions until $(n-1)$-step and depends only on S_n. This property is called **Markov property**. Markov process is a stochastic process with Markov property.

Definition 3.2 (**Transition probability**). The function p defined on $S \times \mathcal{B}$ satisfying

- for any $x \in S$, the function $p(x, \cdot)$ is a probability on $\mathcal{B}$.
- for any $A \in \mathcal{B}$, the function $p(\cdot, A)$ is a Borel function on S.

is called **transition probability**.

Under discrete time setting, Markov process $(X_n)_{n\in\mathbb{Z}_+}$ satisfies

$$\mathbb{P}(X_{n+1} \in A \mid X_0, X_1, \cdots, X_n) = p^{(n)}(x_n, A) \quad \text{a.s.} \tag{3.2}$$

If the transition probability is uniform in time, i.e. $p^{(n)} = p$, then the Markov process is called **uniform** or **time-homogeneous**.

In the case that S is not countable, we have a view from functional analysis. For any measurable function $f\colon S \to \mathbb{R}_+$, Markov property implies that

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = (p_{t-s}f)(X_s) \tag{3.3}$$

where $p_{t-s}(x, dy)$ is a conditional probability distribution of X_t given the information X_s, and

$$(P_{t-s}f)(x) = \int_S p_{t-s}(x, dy)\, f(y) \tag{3.4}$$

In this sense, $p_{t-s}(x, dy)$ is also called **transition kernel**. Roughly speaking, the transition probability $p_{t-s}(x, dy)$ shows the probability that the process moves from x at time s to y at t, where dy in the notation because of the range of possible y. Thus, it will be easy to see

$$\begin{aligned}&(P_{t-s}f)(x) = \\ &\qquad \text{the walk transports the quantity } f \text{ from } x \text{ at time } s \text{ to somewhere in } S \text{ at time } t\end{aligned} \tag{3.5}$$

from the understanding of integral in the right-hand side of (3.4).

Example 3.3 (d-dimensional random walk)**.** Let ξ_k $(k = 1, 2, \cdots)$ be a sequence of i.i.d. d-dimensional random variables with distribution ν. The d-dimensional random walk S starting at $x \in \mathbb{R}^d$ is given by

$$S_0 = x, \qquad S_n = x + \sum_{k=1}^{n} \xi_k$$

Let $\sigma_n(S) = \sigma(S_0, S_1, \cdots, S_n)$. Then since

$$\mathbb{P}(S_{n+1} \in A \mid \sigma_n(S)) = \mathbb{P}(S_n + \xi_{n+1} \in A \mid \sigma_n(S)) = \nu(A - S_n),$$

random walk S is uniform Markov process with transition probability

$$p(x, A) = \nu(A - S_n)$$

which shows that the transition of random walk is *translation invariant* (or *shift invariant*). □

Example 3.4 (absorbing/reflection 1D-random walk)**.** We consider 1-dimensional simple random walk on $\{0, 1, 2, \cdots\}$ with absorbing boundary at 0. Then the transition probability can be written as

$$p(i, i+1) = p, \qquad p(i, i-1) = 1 - p \qquad (i = 1, 2, \cdots), \qquad p(0, 0) = 1 \tag{3.6}$$

If we consider $+1$ (-1) means earn (lose) 1 dollar, then being absorbed at 0 stands for "ruin". The model with the absorbing boundary is called (gambler's) ruin model.

If we assume another boundary condition: $p(0, 1) = 1$, then we call the random walk with "reflection" boundary. □

3.1 n-step transition probability

Now we consider a product set $B = A_1 \times A_2 \times \cdots \times A_n$. We set

$$\begin{aligned} Q(x, B) := \int_{A_1} p^{(0)}(x, dy_1) \int_{A_2} p^{(1)}(y_1, dy_2) \int_{A_3} \cdots \\ \times \int_{A_{n-2}} p^{(n-3)}(y_{n-3}, dy_{n-2}) \int_{A_{n-1}} p^{(n-2)}(y_{n-2}, dy_{n-1})\, p^{(n-1)}(y_{n-1}, A_n) \end{aligned} \tag{3.7}$$

We can imagine that Q shows the weight for the n-step path $(x, y_1, y_2, \cdots, y_{n-3}, y_{n-2}, y_{n-1}, y_n)$. We notice that the initial point x is deterministically fixed but y_1 can be taken from A_1, y_2 from A_2, $\cdots$ and y_n from A_n. The n-th step transition probability $p^{(n-1)}(y_{n-1}, A_n)$ is a Borel function of y_{n-1} and the integral

$$\int_{A_{n-1}} p^{(n-2)}(y_{n-2}, dy_{n-1})\, p^{(n-1)}(y_{n-1}, A_n) \tag{3.8}$$

is a Borel function of y_{n-2}. Therefore, $Q(x, B)$ is a Borel function of x. In general, for $B = S \times S \times \cdots \times S \times A$,

$$Q(x, B) = \int_S \cdots \int_S p^{(0)}(x, dy_1)\, p^{(1)}(y_1, dy_2) \cdots p^{(n-1)}(y_{n-1}, A) = p^{(0)} * p^{(1)} * \cdots p^{(n-1)}(x, A) \tag{3.9}$$

where $f * g$ is the convolution of f and g:

$$(f * g)(x) = \int f(x, z)\, g(z, y)\, dz \tag{3.10}$$

In particular, for the homogeneous setting, the n-step transition probability can be expressed by the n-fold convolution of 1-step transition probabilities.

Remark 3.5. $p^{(0)}(x, dy_1)$ can be understood as $f(x, y_1)dy_1$, where $f(x, y_1)$ is the density function. One might use the definition of convolution by

$$(f * g)(x) = \int f(x - y)\, g(y)\, dy. \tag{3.11}$$

We can consider "shift invariance"

$$p^{(0)}(x, y_1) = p^{(0)}(0, y_1 - x) =: p^{(0)}(y_1 - x) \tag{3.12}$$

Proposition 3.6. *Let $(p^{(n)})_{n\in\mathbb{Z}_+}$ be transition probabilities. Then the following are holds almost surely.*

$$\begin{aligned}
&\mathbb{P}(X_{n+1}\in A_1, X_{n+2}\in A_2, \cdots, X_{n+m}\in A_m \mid \sigma_n(\mathbf{X}))\\
&= \mathbb{P}(X_{n+1}\in A_1, X_{n+2}\in A_2, \cdots, X_{n+m}\in A_m \mid X_n)\\
&= \int_S \cdots \int_S p^{(n)}(X_n, dy_1)\, p^{(n+1)}(y_1, dy_2)\, \cdots\, p^{(n+m-1)}(y_{m-1}, dy_m)\\
&\qquad\qquad \times \mathbb{1}_{\{y_1\in A_1\}}\, \mathbb{1}_{\{y_2\in A_2\}}\, \cdots\, \mathbb{1}_{\{y_m\in A_m\}}\\
&= \int_{A_1} \cdots \int_{A_m} p^{(n)}(X_n, dy_1)\, p^{(n+1)}(y_1, dy_2)\, \cdots\, p^{(n+m-1)}(y_{m-1}, dy_m)
\end{aligned} \tag{3.13}$$

and

$$\begin{aligned}
&\mathbb{E}[f(X_{n+1}, X_{n+2}, \cdots, X_{n+m}) \mid \sigma(X_n)]\\
&= \mathbb{E}[f(X_{n+1}, X_{n+2}, \cdots, X_{n+m}) \mid X_n)\\
&= \int_S \cdots \int_S p^{(n)}(X_n, dy_1)\, p^{(n+1)}(y_1, dy_2)\, \cdots\, p^{(n+m-1)}(y_{m-1}, dy_m)\, f(y_1, y_2, \cdots, y_m)
\end{aligned} \tag{3.14}$$

where f is a bounded measurable function on S.

Proof. We first note that (3.13) and (3.14) are equivalent from definition of integrals. We prove by induction. For $m = 1$,

$$\mathbb{E}[f(X_{n+1}) \mid \sigma_n(\mathbf{X})] = \int_S p^{(n)}(X_n, dy)\, f(y) = \mathbb{E}[f(X_{n+1}) \mid X_n], \tag{3.15}$$

where $\sigma_n(\mathbf{X}) = \sigma(X_0, X_1, \cdots, X_n)$ and $\mathbb{P}(X_{n+1}\in A \mid \sigma_n(\mathbf{X})) = p^{(n)}(X_n, A)$ from definition. This shows (3.14) for $m = 1$. For $m = k$, we assume (3.13) and (3.14). To simplify the notation, we write $f_i = \mathbb{1}_{A_i}$. Since

$$\mathbb{1}_{\{X_{n+1}\in A_1, \cdots, X_{n+k}\in A_k\}} = \prod_{j=1}^{k} \mathbb{1}_{A_j}(X_{n+j}), \tag{3.16}$$

we have for $m = k + 1$,

$$\begin{aligned}
&\mathbb{P}(X_{n+1}\in A_1, X_{n+2}\in A_2, \cdots, X_{n+k+1}\in A_{k+1} \mid \sigma(X_n))\\
&= \mathbb{E}\left[\prod_{j=1}^{k+1} f_j(X_{n+j}) \middle| \sigma_n(\mathbf{X})\right]\\
&= \mathbb{E}\left[\mathbb{E}[f_{k+1}(X_{n+k+1}) \mid \sigma_{n+k}(\mathbf{X})] \prod_{j=1}^{k} f_j(X_{n+j}) \middle| \sigma_n(\mathbf{X})\right]\\
&= \mathbb{E}\left[g(X_{n+k}) \prod_{j=1}^{k} f_j(X_{n+j}) \middle| \sigma_n(\mathbf{X})\right],
\end{aligned} \tag{3.17}$$

where

$$g(x) = \int_S p^{(n+k)}(x, dy)\, f_{k+1}(y). \tag{3.18}$$

By the assumption, we have

$$\begin{aligned}\mathbb{E}[\mathbb{1}_{\{X_{n+1}\in A_1,\cdots,X_{n+k}\in A_k\}} \mid \sigma_n(\mathbf{X})] &= \mathbb{E}\left[\prod_{j=1}^{k}\mathbb{1}_{A_j}(X_{n+j})\middle|\sigma_n(\mathbf{X})\right] \\ &= \int_S\cdots\int_S p^{(n)}(X_n,dy_1)\cdots p^{(n+k-1)}(y_{k-1},dy_k)\prod_{j=1}^{k} f_j(y_j)\end{aligned} \tag{3.19}$$

Substituting into (3.17), we obtain

$$\begin{aligned}&\mathbb{P}(X_{n+1}\in A_1, X_{n+2}\in A_2,\cdots,X_{n+k+1}\in A_{k+1}\mid\sigma(X_n)) \\ &= \int_S\cdots\int_S p^{(n)}(X_n,dy_1)\,p^{(n+1)}(y_1,dy_2)\cdots p^{(n+k-1)}(y_{k-1},dy_k)\prod_{j=1}^{k} f_j(y_j)\,g(y_k) \\ &= \int_S\cdots\int_S\int_S p^{(n)}(X_n,dy_1)\cdots p^{(n+k-1)}(y_{k-1},dy_k)\,p^{(n+k)}(y_k,dy_k)\prod_{j=1}^{k+1} f_j(y_j),\end{aligned} \tag{3.20}$$

which shows the case for $m = k+1$. As the general argument in Lebesgue integral, we can extend from indicator function to bounded measurable functions via approximation by simple functions. ■

Now we denote by

$$\mathbb{P}_x(\cdot) := \mathbb{P}(\cdot \mid X_0 = x) \tag{3.21}$$

the (conditional) probability with initial point $x \in S$. Then we can write the distribution of $\mathbf{X}$ with the initial distribution of X_0 denoted by μ,

$$\begin{aligned}&\mathbb{P}(X_0\in A_0, X_1\in A_1, X_2\in A_2,\cdots,X_m\in A_m) \\ &= \int_{A_0}\int_{A_1}\cdots\int_{A_m}\mu(dx_0)\,p^{(0)}(x_0,dx_1)\,p^{(1)}(x_1,dx_2)\cdots p^{(m-1)}(x_{m-1},dx_m)\end{aligned} \tag{3.22}$$

We denote by $p(m,x;n,A)$ the transition probability from x at mth-step to A in n steps:

$$\begin{aligned}&p(m,x;n,A) \\ &= \int_S\int_S\cdots\int_S p^{(m)}(x,dy_1)\,p^{(m+1)}(y_1,dy_2)\cdots p^{(n-1)}(y_{n-1-m},A) \\ &= p^{(m)} * p^{(m+1)} * \cdots p^{(n-1)}(x,A)\end{aligned} \tag{3.23}$$

By Proposition 3.6,

$$\mathbb{P}(X_n\in A\mid\sigma_m(\mathbf{X})) = p(m,X_m;n,A) \qquad \text{a.s.} \tag{3.24}$$

which yields the probability that X_n is in A under the condition $X_m = x$. In particular, the case $m = 0$ gives

$$\mathbb{P}_x(X_n\in A) = \int_S \mu(dx)\,p(0,x;n,A). \tag{3.25}$$

Chapman–Kolmogorov equation One may see that a Markov process can be decomposed into two or more intervals. For $\ell < m < n$, by Markov property,

$$\begin{aligned}\mathbb{P}(X_n\in A\mid\sigma_\ell(\mathbf{X})) &= \mathbb{E}[\mathbb{P}(X_n\in A\mid\sigma_m(\mathbf{X}))\mid\sigma_\ell(\mathbf{X})] \\ &= \mathbb{E}[p(m,X_m;n,A)\mid\sigma_\ell(\mathbf{X})] \\ &= \int_S p(\ell,X_\ell;m,dy)\;p(m,y;n,A) \qquad \text{a.s.}\end{aligned} \tag{3.26}$$

which leads to the following theorem.

Theorem 3.7 (Chapman–Kolmogorov equation)**.** *For $k < m < n$, we have*

$$p(k, x; n, A) = \int_S p(k, x; m, dy)\, p(m, y; n, A) \tag{3.27}$$

This equation implies that the process moves to A at time n is equivalent to that it moves first to y at some time m and then transits to A at n, where the integral comes from the fact that y can be freely chosen from S.

In the case that p_n is homogeneous in time, the convolution in (3.23) can be written

$$\underbrace{p * p * \cdots * p}_{(n-m)\text{-fold}}(x, A) = p^{*(n-m)}(x, A), \tag{3.28}$$

which is the $(n-m)$-fold convolution. Then the Chapman–Kolmogorov equation is

$$p^{*(n+m)}(x, A) = \int_S p^{*n}(x, dy)\, p^{*m}(y, A) \tag{3.29}$$

If S is countable, integral becomes summation

$$p(m, a; n, b) = \sum_{z \in S} p(m, a; k, z)\, p(k, z; n, b) \tag{3.30}$$

Especially, in time-homogeneous case, $p(m, x; m+1, y)$ does not depends on m, so we abbreviate by $p(x, y)$. The matrix consisting of $p(x, y)$: $\mathbf{P} := (p(x, y)_{x,y \in S}$ is called **transition matrix** of $\mathbf{X}$. Obviously, each element is nonnegative and for $x \in S$, $\sum_{y \in S} p(x, y) = 1$. The Chapman–Kolmogorov equation is expressed as

$$\mathbf{P}^{n+m}(a, b) = \sum_{z \in S} \mathbf{P}^n(a, z)\, \mathbf{P}^m(z, b). \tag{3.31}$$

3.2 Strong Markov property

In the theory of Markov process, even if we can change n to general stopping time, Markov property and Proposition 3.6 hold, which is called *strong Markov property.*

Proposition 3.8 (Strong Markov property)**.** Let τ be a stopping time and $\{p^{(n)}\}_{n \in \mathbb{Z}_+}$ be transition probabilities of $\mathbf{X}$. Then we have the following:

(1)
$$\mathbb{P}(X_{\tau+1} \in A \mid \sigma_\tau(\mathbf{X})) = \mathbb{P}(X_{\tau+1} \in A \mid \tau, X_\tau) = p^{(\tau)}(X_\tau, A) \quad \text{a.s.} \tag{3.32}$$

(2)
$$\mathbb{P}(X_{\tau+1} \in A_1, \cdots X_{\tau+k} \in A_k \mid \sigma_\tau(\mathbf{X})) = \mathbb{P}(X_{\tau+1} \in A_1, \cdots X_{\tau+k} \in A_k \mid \tau, X_\tau) \quad \text{a.s.} \tag{3.33}$$

(3) For bounded Borel function f,
$$\mathbb{E}[f(X_{\tau+1}, \cdots, X_{\tau+k}) \mid \sigma_\tau(\mathbf{X})] = \mathbb{E}[f(X_{\tau+1}, \cdots, X_{\tau+k}) \mid \tau, X_\tau] \quad \text{a.s.} \tag{3.34}$$

Before the proof, we consider the meaning of the statements. The information (τ, X_τ) gives when and where the process stopped. Since $\sigma(\tau, X_\tau) \subset \sigma_\tau(\mathbf{X})$, the statement

$$\mathbb{P}(X_{\tau+1} \in A \mid \sigma_\tau(\mathbf{X})) = \mathbb{P}(X_{\tau+1} \in A \mid \tau, X_\tau) \tag{3.35}$$

means that $X_{\tau+1}$ is just determined by the information τ and X_τ. We do not need whole information of the process $\sigma_\tau(\mathbf{X})$, which is similar to the Markov property.

Proof. We start with the proof of (3). Since $\sigma_\tau(\mathbf{X}) = \sigma(X_0, X_1, \cdots, X_\tau)$, for $\Lambda \in \sigma_\tau(\mathbf{X})$,

$$\Lambda_n := \Lambda \cap \{\tau = n\} \in \sigma_n(\mathbf{X}) \tag{3.36}$$

Thus

$$\begin{aligned}
\mathbb{E}[f(X_{\tau+1}, \cdots, X_{\tau+\ell})\mathbb{1}_\Lambda] &= \sum_{n=0}^{\infty} \mathbb{E}\Big[f(X_{n+1}, \cdots, X_{n+\ell})\mathbb{1}_{\Lambda_n}\Big] \\
&= \sum_{n=0}^{\infty} \mathbb{E}\Big[\mathbb{E}[f(X_{n+1}, \cdots, X_{n+\ell}) \mid \sigma_n(\mathbf{X})]\mathbb{1}_{\Lambda_n}\Big] \\
&= \sum_{n=0}^{\infty} \mathbb{E}\Big[\mathbb{E}[f(X_{n+1}, \cdots, X_{n+\ell}) \mid X_n]\mathbb{1}_\Lambda \mathbb{1}_{\{\tau=n\}}\Big] \\
&= \sum_{n=0}^{\infty} \mathbb{E}\Big[\mathbb{E}[f(X_{\tau+1}, \cdots, X_{\tau+\ell}) \mid X_\tau]\mathbb{1}_\Lambda \mathbb{1}_{\{\tau=n\}}\Big] \\
&= \mathbb{E}\Big[\mathbb{E}[f(X_{\tau+1}, \cdots, X_{\tau+\ell}) \mid X_\tau]\mathbb{1}_\Lambda\Big],
\end{aligned} \tag{3.37}$$

which yields (3).

To prove (2), we take

$$f(x_1, \cdots, x_\ell) = \prod_{k=1}^{\ell} \mathbb{1}_{A_k}(x_k). \tag{3.38}$$

Then (3.37) implies (2).

To prove (1), we take

$$\ell = 1, \qquad f(x) = \mathbb{1}_A(x). \tag{3.39}$$

Thus we have

$$\mathbb{E}[\mathbb{1}_A(X_{n+1}) \mid X_n] = p^{(n)}(X_n, A). \tag{3.40}$$

Therefore, (3.37) implies that

$$\mathbb{P}(X_{\tau+1} \in A \mid \sigma_\tau(\mathbf{X})) = p^{(\tau)}(X_\tau, A) \quad \text{a.s.} \tag{3.41}$$

■

In particular, when $\mathbf{X}$ is a uniform Markov process, (1) in Lemma 3.8 is

$$\mathbb{P}(X_{\tau+1} \in A \mid \sigma_\tau(\mathbf{X})) = p(X_\tau, A) \quad \text{a.s.} \tag{3.42}$$

Then the condition (τ, X_τ) can be reduce to only X_τ.

Example 3.9 (Strong Markov property)**.** Let $\mathbf{X}$ be a time-homogeneous Markov process and τ be the first hitting time at $a \in S$, i.e. $\tau = \inf\{n \geq 1 \colon X_n = a\}$, where we set $\inf \varnothing = \infty$ and $\mathbb{P}(\tau < \infty) = 1$.

Then $\tau + n$ is also a stopping time and

$$\mathbb{P}(X_{\tau+n+1} \in A \mid \sigma_{\tau+n}(\mathbf{X})) = p(X_{\tau+n}, A) \qquad \text{a.s.} \tag{3.43}$$

We denote by $\mathbf{X}^+ = \{X_k^+ := X_{\tau+k},\ k \in \mathbb{Z}_+\}$ the process after τ. Also we have

$$\mathbb{P}(X_{n+1}^+ \in A \mid \sigma_n(\mathbf{X}^+)) = \mathbb{E}\left[\mathbb{E}\left[\mathbb{1}_{\{X_n^+ \in A\}} \mid \sigma_{\tau+n}(\mathbf{X})\right] \mid \sigma_n(\mathbf{X}^+)\right] = p(X_n^+, A) \quad \text{a.s.} \tag{3.44}$$

which shows that $\mathbf{X}^+$ is also time-homogeneous Markov process with the same transition probability p as $\mathbf{X}$. By $X_\tau = a$,

$$\mathbb{E}[f(X_1^+, \cdots X_k^+) \mid \sigma_\tau(\mathbf{X})) = \mathbb{E}_a[f(X_1^+, \cdots, X_k^+)] \tag{3.45}$$

which yields that the processes $\mathbf{X}^+$ and $\mathbf{X}$ are independent. Therefore, we can treat the process after τ independent of the process until τ. □

3.3 "Discrete time" Markov chain

We consider time-homogeneous Markov process through this section. If the state space is countable, the Markov process is called **Markov chain**. We consider $\mathbf{X}$ is a Markov chain and its transition matrix $\mathbf{P} = (p(x,y))_{x,y\in S}$.

Remark 3.10. This definition of Markov chain is actually old-fashioned. In general, even if S is uncountable, the discrete-time Markov process is called Markov chain. Thus the word "discrete time Markov chain" sounds a bit weird.

Transition kernel and Markov operator The transition matrix can no longer defined for uncountable S. Thus we consider the *transition kernel* $p(x, dy)$. By the definition of transition kernel, we have the expression

$$\mathbb{E}[f(X_{n+1}) \mid X_n = x] = \int_S p(x,dy)\, f(y) \tag{3.46}$$

and define **Markov operator P** by

$$(\mathbf{P}f)(x) := \int_S p(x,dy)\, f(y), \qquad f \in L^\infty. \tag{3.47}$$

For distribution μ (distribution of X_0), we can also write

$$(\mu\mathbf{P})(A) := \int_S \mu(dx)\, p(x,A) \tag{3.48}$$

and $\mu\mathbf{P}$ is the distribution of X_1. Indeed,

$$\int_S \mu(dx)\, p(x,A) = \int_S \mathbb{P}(X_1 \in A \mid X_0 = x)\, \mu(dx) = \mathbb{P}(X_1 \in A) \tag{3.49}$$

Therefore, instead of the sequence $(X_0, X_1, X_2, \cdots)$, the following sequence

$$(\mu, \mu\mathbf{P}, \mu\mathbf{P}^2, \cdots) \tag{3.50}$$

also shows the Markov chain (X_n).

Remark 3.11. If $\mu(dx) = \varphi(x)dx$, i.e. if μ is absolutely continuous w.r.t. Lebesgue measure dx, then we can define the probability density function $\varphi(x) \in L^1$. Thus we have the expression

$$\mathbb{E}[f(X)] = \int_S f(x)\, \varphi(x)\, dx = \langle f, \varphi \rangle, \tag{3.51}$$

which is called Banach dual. Roughly saying, $f \in L^\infty$ is the observable and $\varphi \in L^1$ is the weight.

Recurrence and transience We investigate the probability that the Markov chain comes back to the initial point. Let $\tau_x = \inf\{n \geq 1 \colon X_n = x\}$ be the first hitting time at $x \in S$.

Definition 3.12 (Recurrence and transience)**.** *We characterize the state $x \in S$ by the following:*

- $x \in S$ is **recurrent** when $\mathbb{P}_x(\tau_x < \infty) = 1$.
- $x \in S$ is **transient** when $\mathbb{P}_x(\tau_x < \infty) < 1$.
- $x \in S$ is said to be positive recurrent when $\mathbb{E}_x[\tau_x] < \infty$.
- $x \in S$ is said to be null recurrent when $\mathbb{E}_x[\tau_x] = \infty$.

We are interested in the condition for recurrence. We set

$$p_n(x,y) := \mathbb{P}_x(X_n = y), \qquad f_n(x,y) := \mathbb{P}_x(\tau_y = n), \qquad f_0(x,y) := 0 \tag{3.52}$$

and define the generating function

$$P(x,y;s) = \sum_{n=0}^{\infty} p_n(x,y)\, s^n, \qquad F(x,y;s) = \sum_{n=0}^{\infty} f_n(x,y)\, s^n \tag{3.53}$$

Lemma 3.13. *we have the following relations among the transition probability p, the distribution of the first hitting time f, and those generating functions:*

$$p_n(x,y) = \sum_{k=0}^{n} f_k(x,y)\, p_{n-k}(y,y) \tag{3.54}$$

$$P(x,y;s) \begin{cases} = (1 - F(x,x;s))^{-1} & x = y \\ = F(x,y;s)\, P(y,y;s) & x \neq y \end{cases} \tag{3.55}$$

Question 3.14. Prove Lemma 3.13.

Combining Lemma 3.13 with the fact that

$$\mathbb{P}_x(\tau_x < \infty) = \sum_{n=1}^{\infty} f_n(x,x) = \lim_{s \to 1} F(x,x;s) \tag{3.56}$$

we obtain the following theorem

Theorem 3.15 (Criterion for recurrence and transience.)**.** *We have the following criteria:*

1. *The necessary and sufficient condition that x is recurrent is*

$$\sum_{n=0}^{\infty} p_n(x,x) = \infty \tag{3.57}$$

2. *The necessary and sufficient condition that x is transient is*

$$\sum_{n=0}^{\infty} p_n(x,x) < \infty \tag{3.58}$$

3. *If x is transient, then for any y,*

$$\lim_{n \to \infty} p_n(y,x) = 0 \tag{3.59}$$

Proof. The items **1.** and **2.** are trivial from (3.55) and (3.56).

In the case $y = x$, the item **2.** yields the item **3.** directly. Thus we focus on the case $y \neq x$. For any $\varepsilon > 0$, we choose $N = N(\varepsilon)$ such that $\sum_{n=N}^{\infty} p_n(x,x) < \varepsilon$ by the item **2.**. Then, by (3.54),

$$\begin{aligned} p_n(y,x) &= \sum_{k=0}^{n} f_k(y,x)\, p_{n-k}(x,x) \\ &\leq \sum_{k=0}^{\ell} p_{n-k}(x,x) + \sum_{k=\ell}^{\infty} f_k(y,x) \end{aligned} \tag{3.60}$$

Now we fix $\ell = \ell(\varepsilon)$ such that $\sum_{k=\ell}^{\infty} f_k(y,x) < \varepsilon$. Therefore, if we take $n > \ell + N$, then $p_n(y,x) < 2\varepsilon$. Since $\varepsilon > 0$ is arbitrary, we obtain the item **3.**. ■

Example 3.16 (Simple random walk on $\mathbb{Z}$)**.** Let

$$p(x, x+1) = p, \qquad p(x, x-1) = 1 - p \tag{3.61}$$

Then we have

$$p_{2k}(x,x) = \binom{2k}{k} p^k (1-p)^k, \qquad k = 1, 2, \cdots \tag{3.62}$$

By Stirling's formula: $n! \sim \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n}$, for k large enough,

$$p_{2k}(x,x) \sim \frac{(4p(1-p))^k}{\sqrt{k\pi}} \tag{3.63}$$

Therefore, the random walk is recurrent for $p = \frac{1}{2}$ and transient for $p \neq \frac{1}{2}$. □

When x is recurrent, the Markov processes before and after τ_x denoting $\mathbf{X}^- = \{X_t\}_{t<\tau_x}$ and $\mathbf{X}^+ = \{X_t\}_{t\leq\tau_x}$ are independent and identically distributed. Therefore x is recurrent for $\mathbf{X}^+$, which means that $\mathbf{X} = \{X_t\}_{t\in\mathbb{Z}_+}$ returns to x more than twice with probability 1. Repeatedly, $\mathbf{X}$ returns to x infinitely many times with probability 1.

Irreducibility When the Markov process $\mathbf{X}$ satisfies

$$\mathbb{P}_x(X_m = y) > 0 \tag{3.64}$$

for some $m \geq 1$, x **is accessible to** y denoting by $x \to y$. If for any x and y, we have $x \to y$ and $y \to x$, then the Markov process $\mathbf{X}$ is said to be **irreducible**. The random walk with absorbing boundary is not irreducible because the process cannot transit from 0.

Theorem 3.17. *If x is recurrent and $x \to y$, then*

$$\mathbb{P}_y(\tau_x < \infty) = 1 \tag{3.65}$$

Proof. By $x \to y$, for some $m \geq 1$,

$$\mathbb{P}_x(X_1 \neq y, \cdots, X_{m-1} \neq y, X_m = y) > 0 \tag{3.66}$$

Since x is recurrent, the probability that x can be returned infinitely many times is 1. Therefore, setting

$$\tau_x(m) := \inf\{k \geq m+1 \colon X_k = x\} \tag{3.67}$$

we have

$$\mathbb{P}_x(\tau_x(m) < \infty) = 1. \tag{3.68}$$

Thus

$$\begin{aligned} 0 &= \mathbb{P}_x(\tau_x(m) = \infty) \\ &\geq \mathbb{P}_x(X_1 \neq y, \cdots, X_{m-1} \neq y, X_m = y, \tau_x(m) = \infty) \\ &= \mathbb{E}_x[\mathbb{P}(\tau_x(m) = \infty \mid \sigma_m(\mathbf{X}))\mathbb{1}_{\{X_1\neq y,\cdots,X_{m-1}\neq y, X_m = y\}}] \\ &= \mathbb{P}_y(\tau_x = \infty)\, \mathbb{P}_x(X_1 \neq y, \cdots, X_{m-1} \neq y, X_m = y) \end{aligned} \tag{3.69}$$

which implies that $\mathbb{P}_y(\tau_x = \infty) = 0$. Hence we conclude

$$\mathbb{P}_y(\tau_x < \infty) = 1 \tag{3.70}$$

■

Theorem 3.18. *If* $\mathbf{X}$ *is irreducible, then each of the following holds:*

(1) *all points are recurrent*

(2) *all points are transient*

In the case (1), the process $\mathbf{X}$ *is said to be* **recurrent**, *in the case (2)* **transient**.

Proof. Let x be recurrent. For any y we choose $n, m \geq 1$ and $a > 0$ such that

$$\mathbb{P}_x(X_n = y) \geq a, \qquad \mathbb{P}_y(X_m = x) \geq a \tag{3.71}$$

Thus, by Chapman–Kolmogorov equation,

$$\mathbb{P}_y(X_{m+k+n} = y) \geq \mathbb{P}_y(X_m = x)\,\mathbb{P}_x(X_k = x)\,\mathbb{P}_x(X_n = y) \geq a^2 \mathbb{P}_x(X_k = x) \tag{3.72}$$

Taking summation over k, we have

$$\sum_{\ell=0}^{\infty} p_\ell(y, y) \geq \sum_{k=0}^{\infty} p_{m+k+n}(y, y) \geq a^2 \sum_{k=0}^{\infty} p_k(x, x) = \infty \tag{3.73}$$

By the item **1.** in Theorem 3.15, y is recurrent. Therefore, if there exists a recurrent point, then all points are recurrent, and (1) holds. ■

Combining Theorem 3.17, we have the following corollary.

Corollary 3.19. *If irreducible* $\mathbf{X}$ *has a recurrent point, all points are recurrent and any two points are transitive with probability 1.*

Example 3.20. Let $(\xi_k)_{k \in \mathbb{N}}$ be i.i.d. with

$$\nu_x := \mathbb{P}(\xi_1 = x), \quad \mathbb{E}[\xi_1] = 0, \quad v := \mathbb{V}\mathrm{ar}(\xi_1) \in (0, \infty) \tag{3.74}$$

Since $p(x, y) = \nu_{y-x}$, we have $p_n(x, x) = p_n(0, 0)$. Then

$$\begin{aligned} G_n(0, x) &= \sum_{k=0}^{n} p_k(0, x) \\ &= \sum_{k=0}^{n} \left(\sum_{\ell=0}^{k} f_\ell(0, x) p_{k-\ell}(x, x) \right) \\ &= \sum_{\ell=0}^{n} f_\ell(0, x) \sum_{k=\ell}^{n} p_{k-\ell}(0, 0) \\ &\leq G_n(0, 0) \end{aligned} \tag{3.75}$$

Thus

$$G_n(0, 0) \geq \frac{1}{2M+1} \sum_{|x| \leq M} G_n(0, x) \geq \frac{1}{2M+1} \sum_{k=0}^{n} \sum_{|x| \leq \frac{k}{n} M} p_k(0, x) \tag{3.76}$$

We take $M = cn$. Then

$$\begin{aligned} G_n(0, 0) &\geq \frac{1}{2cn+1} \sum_{k=0}^{n} \mathbb{P}_0(|X_k| \leq kc) \\ &= \frac{1}{2cn+1} \sum_{k=0}^{n} \Big(1 - \mathbb{P}_0(|X_k| > kc) \Big) \\ &\geq \frac{1}{2cn+1} \sum_{k=1}^{n} \Big(1 - \frac{v}{kc^2} \Big) \end{aligned} \tag{3.77}$$

where the last inequality we apply Chebyshev's inequality. Therefore, as $n \to \infty$, we have

$$\sum_{n=0}^{\infty} p_n(0,0) = \lim_{n\to\infty} G_n(0,0) \geq \lim_{n\to\infty} \frac{1}{2cn+1}\Big(n - \frac{v}{c^2}\log n\Big) = \frac{1}{2c} \tag{3.78}$$

Hence, as $c \to 0$, we conclude that 0 is recurrent by Theorem 3.15. □

Question 3.21 (Recurrence and transience for d-dimensional SRW)**.** Let $d \in \mathbb{N}$, $e_1 = (1,0,\cdots,0), \cdots, e_d = (0,\cdots,0,1)$ be d-dimensional unit vectors, and $(\xi_k)_{k\in\mathbb{N}}$ be d-dimensional i.i.d. random variables with

$$\mathbb{P}(\xi_1 = e_j) = \frac{1}{2d}, \quad \mathbb{P}(\xi_1 = -e_j) = \frac{1}{2d}. \tag{3.79}$$

We define d-dimensional SRW starting at $\mathbf{0}$ by

$$X_0 = \mathbf{0} \in \mathbb{Z}^d, \qquad X_n = \mathbf{0} + \sum_{k=1}^{n} \xi_k \tag{3.80}$$

Show that

- $d = 2 \Rightarrow \mathbf{0}$ is recurrent
- $d \geq 3 \Rightarrow \mathbf{0}$ is transient

3.4 Invariant/stationary distribution

The rest part of this chapter is advanced for the course, so if it is too difficult you can skip so far. However, when we want to know the asymptotic behavior of Markov processes, the invariant distribution is one of the main analysis. At least check the definition and simple examples.

Since the state space S is finite, so we assume labeling $S = \{1,2,\cdots,s$ where $s \in \mathbb{N}$. Then we have

$$\begin{aligned}\mathbb{P}(X_{n+1} = j) &= \sum_{i=1}^{s} \mathbb{P}(X_n = i \mid X_n = i)\mathbb{P}(X_n = i) \\ &= \sum_{i=1}^{s} \mathbb{P}(X_n = i)\, p(i,j), \qquad 1 \leq j \leq s,\ n = 0,1,\cdots\end{aligned} \tag{3.81}$$

Then we define the vector

$$\mu_n := (\mathbb{P}(X_n = 1), \mathbb{P}(X_n = 2), \cdots, \mathbb{P}(X_n = s)) \tag{3.82}$$

which is the distribution of X_n. We can write with the transition matrix $\mathbf{P} = (p(i,j))_{1\leq i,j\leq s}$

$$\mu_{n+1} = \mu_n\, \mathbf{P}, \qquad n = 0,1,\cdots \tag{3.83}$$

where $\mu_0 = \mu$ is the initial distribution. Repeatedly,

$$\mu_n = \mu\, \mathbf{P}^n \tag{3.84}$$

Therefore, the computation of the probability $\mathbb{P}(X_n = i)$ can be reduced to the computation of $\mathbf{P}^n$ (diagonalization or Jordan form). When we consider the limit

$$\lim_{n\to\infty} \mathbb{P}(X_n = i) \tag{3.85}$$

and assume that for all $i \in S$ the limit exists. Then we have

$$\lim_{n\to\infty} \mu_n =: \widetilde{\pi}, \qquad \widetilde{\pi} = \widetilde{\pi}\, \mathbf{P} \tag{3.86}$$

which means that $\widetilde{\pi}$ is the fixed point of the operation: multiply $\mathbf{P}$ from the right. We call this $\widetilde{\pi}$ *stationary* or *invariant distribution* defined as follows.

Definition 3.22. When the probability distribution $\Pi = \{\pi(x)\}_{x\in S}$ for the transition probability $p(x, y)$ satisfies for any $y \in S$,

$$\pi(y) = \sum_{x\in S} \pi(x)\, p(x, y) \tag{3.87}$$

then Π is called **invariant distribution**.

(3.87) can be written as

$$\Pi = \Pi\, \mathbf{P} \tag{3.88}$$

By Chapman–Kolmogorov equation, we have

$$\pi(z) = \sum_{y\in S} \pi(y)\, p(y, z) = \sum_{x,y\in S} \pi(x)\, p(x, y)\, p(y, z) = \sum_{x\in S} \pi(x)\, p_2(x, z) \tag{3.89}$$

Repeatedly, we obtain

$$\pi(y) = \sum_{x\in S} \pi(x)\, p_k(x, y), \qquad \Pi = \Pi\, \mathbf{P}^k \tag{3.90}$$

The equation (3.90) means that the initial distribution Π of the process $\mathbf{X}$ is the same as the distribution after time k. Therefore, the process after time k, $\mathbf{X}^+ = \{X_n^+ := X_{n+k}\}_{n\in\mathbb{Z}_+}$ is a Markov chain with the initial distribution Π and transition probability $\mathbf{P}$ has the same probability distribution as $\mathbf{X}$. Hence

$$\text{distribution of } (X_{n_1}, X_{n_2}, \cdots, X_{n_\ell}) = \text{distribution of } (X_{n_1+k}, X_{n_2+k}, \cdots, X_{n_\ell+k}) \tag{3.91}$$

which yields that $\mathbf{X}$ is a stationary Markov chain.

Remark 3.23. We have similar terms invariant "distribution" and "measure". The terms are a bit different:

- Distribution is normalized (sum of the elements is 1)
- Measure is not normalized (∞ is allowed)

Theorem 3.24. *If the transition matrix* $\mathbf{P}$ *is finite size, then the Markov chain* $\mathbf{X}$ *with* $\mathbf{P}$ *has the invariant distribution.*

The proof is not easy, so we will skip the proof of this theorem (actually we have several proofs. We can use Brouwer's fixed point theorem for compact convex set). In the case that the state space is finite, we have the invariant distribution. However, if the transition matrix is infinite size, then the existence of the invariant distribution is not always.

We take several examples.

Example 3.25 (Symmetric simple random walk on $\mathbb{Z}$)**.** The symmetric simple random walk on $\mathbb{Z}$ is a Markov chain with transition probability

$$p(i, i+1) = p(i, i-1) = \frac{1}{2} \tag{3.92}$$

Thus the invariant distribution satisfies the equation

$$\pi(i) = \frac{1}{2}(\pi(i+1) + \pi(i-1)) \quad i \in \mathbb{Z}, \qquad \sum_{i\in\mathbb{Z}} \pi(i) = 1 \tag{3.93}$$

Then

$$a := \pi(i) - \pi(i-1) = \pi(i+1) - \pi(i) \tag{3.94}$$

which yields $\pi(j) = ja + \pi(0)$ $(j \in \mathbb{Z})$. However, $\{\pi(i)\}_{i\in\mathbb{Z}}$ cannot be probability. Hence symmetric simple random walk does not have the invariant distribution. □

Example 3.26 (Invariant distribution of random walk with reflection boundary)**.** We consider a random walk with reflection boundary at 0, The transition probability is

$$p(i, i+1) = p, \quad p(i, i-1) = 1-p, \quad i = 1, 2, \cdots, \qquad p(0, 1) = 1 \tag{3.95}$$

The equation for invariant distribution is

$$\begin{aligned} \pi(j) &= \pi(j-1)p + \pi(j+1)(1-p) \\ \pi(1) &= \pi(0) + \pi(2)(1-p), \quad \pi(0) = \pi(1)(1-p) \end{aligned} \tag{3.96}$$

Thus $\Delta(j) = \pi(j+1) - \pi(j)$

$$\begin{aligned} &\Delta(j)(1-p) = \Delta(j-1)p \\ &\Delta(1)(1-p) = (\pi(2) - \pi(1))(1-p) = \pi(1) - \pi(0) - \pi(1)(1-p) = \left(\frac{p}{1-p} - 1\right)\pi(0) \end{aligned} \tag{3.97}$$

Since

$$\pi(n) = \sum_{j=1}^{n-1} \Delta(j) + \pi(1), \tag{3.98}$$

for $p < \frac{1}{2}$, we can obtain the invariant distribution $\pi(n)$ as

$$\pi(0) = \frac{1-2p}{2(1-p)}, \qquad \pi(n) = \frac{1-2p}{2p(1-p)}\left(\frac{p}{1-p}\right)^n \quad n = 1, 2, \cdots \tag{3.99}$$

□

We sometimes say that "the transition matrix $\mathbf{P}$ is irreducible" when the Markov chain $\mathbf{X}$ with transition matrix $\mathbf{P}$ is irreducible.

Theorem 3.27. *If the finite transition matrix* $\mathbf{P}$ *is irreducible, then the invariant distribution is unique.*

Proof. The proof is done by Perron–Frobenius theorem. ■

Question 3.28. Proof Theorem 3.27.

If the process is irreducible, then the existence of invariant distribution can be proven by the probability and expectation of the hitting times without Brouwer's fixed point theorem.

Example 3.29 (Invariant distribution of random walk with absorbing boundary)**.** We consider the simplest case. Let $S = \{1, 2\}$ and

$$p(1, 1) = 0, \quad p(1, 2) = 1, \quad p(2, 1) = 0, \quad p(2, 2) = 1 \tag{3.100}$$

which implies that the process starts from 1, always moves to 2 and stops there. Therefore, it is not irreducible. However, obviously $\pi = (0, 1)$ is the invariant distribution. □

In general, even if the process is not irreducible, we can determine the invariant distribution. We decompose the state space S into equivalence class by irreducibility $\leftrightarrow$. We define "open" and "closed" by transitive or absorbing subset. In each closed subset, we can define invariant distribution because it is irreducible inside the subset. Denoting by π_i each invariant distribution extended to S by putting 0 for outside states, it is known that the following convex combination can be express all invariant distribution of S:

$$\sum_{i=1}^{N} t_i \, \pi_i \qquad t_i \geq 0, \ \sum_{i=1}^{N} t_i = 1 \tag{3.101}$$

In particular, the necessary and sufficient condition for the unique invariant distribution is that there exists only one closed equivalent class.

Example 3.30. We consider the following transition matrices:

$$P_1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{7}{10} & \frac{3}{10} \\ 0 & 0 & \frac{3}{10} & \frac{7}{10} \end{pmatrix} \qquad P_2 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \tag{3.102}$$

The process generated by transition matrix P_1 is not irreducible. The state space is separated into $\{1, 2\}$ and $\{3, 4\}$. Thus

$$\pi_A = (\tfrac{1}{2}, \tfrac{1}{2}, 0, 0), \quad \pi_B = (0, 0, \tfrac{1}{2}, \tfrac{1}{2}) \tag{3.103}$$

are invariant distributions. Actually $\frac{3}{10}\pi_A + \frac{7}{10}\pi_B$ is also an invariant distribution. If we start from $\{1\}$ or $\{2\}$, then we cannot move to $\{3, 4\}$.

The process generated by P_2 is not irreducible but the invariant distribution is unique. The process can move from $\{1, 2\}$ to $\{3, 4\}$ but the opposite is impossible. Thus the invariant distribution is $\pi = (0, 0, \frac{1}{2}, \frac{1}{2})$. □

The sufficient condition for the unique invariant distribution is that the eigenvalue 1 is **simple**. In the example above P_1 can have two 2×2-blocks and each block has the eigenvalue 1, which implies that the eigenvalue 1 is duplicated, not simple.

Periodicity We consider a Markov chain with transition matrix $\mathbf{P}$. The greatest common divisor of the following set

$$\{n \in \mathbb{N} \colon p_n(x, x) > 0\} \tag{3.104}$$

is called **period** of x for with respect to $\mathbf{P}$. When the process $\mathbf{X}$ is irreducible, all states have the same period. If the period is 1, $\mathbf{P}$ is said to be **aperiodic**.

Theorem 3.31. *Let $\mathbf{X}$ be a Markov chain with transition matrix $\mathbf{P}$. The necessary and sufficient condition that $\mathbf{P}$ is irreducible and aperiodic is that there exists $n \in \mathbb{N}$ such that the all elements of $\mathbf{P}^n$ are positive ($\mathbf{P}$ is called* **primitive**).

It is well-known that the limit $\lim_{n\to\infty} \mu \, \mathbf{P}^n$ is equal to the invariant distribution when $\mathbf{P}$ is primitive. Moreover, the limit does not depend on μ because the invariant distribution is unique by irreducibility. Thus there exists the limit

$$\lim_{n\to\infty} \mathbf{P}^n \tag{3.105}$$

Theorem 3.32. *If the transition matrix $\mathbf{P}$ is irreducible and aperiodic, then for any initial distribution μ the following limit exists*

$$\lim_{n\to\infty} \mu \, \mathbf{P}^n \tag{3.106}$$

Let $\mathbf{1}$ be a vector with all elements 1. Then since $\mathbf{P}\,\mathbf{1} = \mathbf{1}$, the eigenvalue of $\mathbf{P}$ is 1.

Theorem 3.33. *Let $\mathbf{P}$ be finite, irreducible and aperiodic, and π be the invariant distribution of $\mathbf{P}$. Then*

(1) *the multiplicity of eigenvalue 1 is one.*

(2) *the absolute value of other eigenvalue of $\mathbf{P}$ is strictly smaller than 1.*

(3) *let $r \in \mathbb{C}$ be the second eigenvalue. Then there exist a constant $C_1 > 0$ and $c_2 \geq 0$ such that for any initial distribution μ and $n \in \mathbb{N}$,*

$$\|\mu\,\mathbf{P}^n - \pi\| \leq C_1\, n^{c_2}\, r^n \tag{3.107}$$

where $\|v\|$ stands for the length of vector v.

The proofs of (1) and (2) are due to the Perron–Frobenius theorem. The proof of (3) is done by Jordan form.

A Review of linear algebra

Perron–Frobenius theorem Perron proved that the nonnegative square matrix has a unique (dominant) eigenvalue assuming it is primitive, and Frobenius shown under the irreducible setting.

Definition A.1 (Primitivity)**.** The nonnegative matrix A is called **primitive** (or **regular**) if there is k such that for any i, j $A^k_{ij} > 0$.

Definition A.2 (Irreducibility)**.** The nonnegative matrix A is called **irreducible** if for any i, j there is $k = k(i, j)$ such that $A^k_{ij} > 0$.

Example A.3. The left matrix is primitive, however, the matrix on right-hand side is irreducible but not primitive:

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{A.108}$$

□

If A is irreducible then $I + A$ is primitive. Indeed, the binomial expansion

$$(I + A)^k = I + \binom{k}{1} A + \binom{k}{2} A^2 + \cdots \tag{A.109}$$

will eventually have positive elements if k is large enough.

Theorem A.4 (Perron–Frobenius theorem)**.** *Let A be a nonnegative and irreducible matrix. Then we have the following*

- *the dominant eigenvalue λ_{PF} is real, positive with associated nonnegative left and right eigenvectors.*
- *all other eigenvalue λ satisfy $|\lambda| \leq \lambda_{\mathrm{PF}}$*
- *the dominant eigenvalue λ_{PF} is simple and has multiplicity 1.*
- *if A is primitive, then all other eigenvalue λ satisfy $|\lambda| < \lambda_{\mathrm{PF}}$*

Jordan normal form Jordan normal form shows the decomposition of the matrix into the exponential and polynomial decays. The exponential decay corresponds to eigenvalues and polynomial part is related to the block size. Even if the matrix cannot be diagonalized Jordan normal form yields the essential structure of the matrix with the simplest manner.

Definition A.5 (Jordan block)**.** We define k-th **Jordan block** $J(\alpha, k)$ by

$$\begin{pmatrix} \alpha & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & \alpha & 1 & 0 & \cdots & 0 & 0 \\ \vdots & 0 & \alpha & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \ddots & \ddots & \vdots \\ \vdots & \vdots & & & \ddots & \ddots & \vdots \\ \vdots & \vdots & & & & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & \cdots & \cdots & \alpha \end{pmatrix} \tag{A.110}$$

In particular,

$$J(\alpha, 1) = (\alpha), \qquad J(\alpha, 2) = \begin{pmatrix} \alpha & 1 \\ 0 & \alpha \end{pmatrix}, \qquad J(\alpha, 3) = \begin{pmatrix} \alpha & 1 & 0 \\ 0 & \alpha & 1 \\ 0 & 0 & \alpha \end{pmatrix} \tag{A.111}$$

Definition A.6 (Jordan matrix)**.** The direct sum of Jordan blocks is called **Jordan matrix**:

$$J = \begin{pmatrix} J_1 & & & \\ & J_2 & & \mathbf{0} \\ & & J_3 & \\ & \mathbf{0} & & \ddots \\ & & & & J_s \end{pmatrix} \tag{A.112}$$

where $J_i = J(\alpha, k_i)$.

Theorem A.7. *For any square matrix A is similar to the unique Jordan matrix J.*

Example A.8. Let

$$A = \begin{pmatrix} 6 & -3 & -2 \\ 4 & -1 & -2 \\ 3 & -2 & 0 \end{pmatrix} \tag{A.113}$$

Then the characteristic equation $\Phi_A(x)$ is

$$\Phi_A(x) = (x-1)(x-2)^2 \tag{A.114}$$

We solve $(A - I)\mathbf{x} = \mathbf{0}$ and $(A - 2I)\mathbf{x} = \mathbf{0}$. Then we obtain

$$\mathbf{p}_1 = (1, 1, 1)^\mathsf{T}, \qquad \mathbf{p}_2 = (2, 2, 1)^\mathsf{T} \tag{A.115}$$

Therefore, the Jordan matrix is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} \tag{A.116}$$

Solve $(A - 2I)\mathbf{x} = \mathbf{p}_2$ and obtain $\mathbf{p}_3 = (1, 0, 1)^\mathsf{T}$. Thus

$$P = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix} \tag{A.117}$$

Therefore,

$$A = PJP^{-1} \tag{A.118}$$

□

Once we have the expression $A = PJP^{-1}$, we obtain

$$A^n = PJ^nP^{-1} \tag{A.119}$$

and for each Jordan block $J_k(\lambda_k)$ of size k with eigenvalue λ_k

$$J_k(\lambda_k)^n = \begin{pmatrix} \lambda_k^n & \binom{n}{1}\lambda_k^{n-1} & \binom{n}{2}\lambda^{n-2} & \cdots \\ 0 & \lambda_k^n & \binom{n}{1}\lambda_k^{n-1} & \cdots \\ \vdots & & \ddots & \vdots \end{pmatrix} \tag{A.120}$$

Let λ_2 is the second largest eigenvalue ($|\lambda_2| < \lambda_{\mathsf{PF}}$). If the matrix A is diagonalizable, we have exponential decay with the speed $O(|\lambda_2|^n)$. If it is not diagonalizable, i.e. it has Jordan block, then it has the term $\binom{n}{k-1}\lambda^{n-(k-1)}$ which yields the decay with the speed $n^{k-1}|\lambda_2|^n$.